

Zane Beeai

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Education

University of Waterloo, B.A.Sc. in Biomedical Engineering, Minor in Computer Science Class of 2030

- **Coursework:** Calculus I & II, Multivariable Calculus, Linear Algebra, Differential Equations, Physics I & II
- **Testing:** SAT: 800M/720R, CCC Honour Roll ×2, USACO Gold

Experience

Co-Founder, Contractual (VC-backed startup) – Toronto, Ontario Jan 2025 – Present

- B2G software enabling small contractors to find and win public tenders with guided AI tooling
- Helped contractors score **\$600,000+** in Canadian government contracts through improved contract-bidding UX
- Built RAG-based backend using Pinecone, LangChain, and custom chunking for NLP-driven tender analysis
- Led development of matching engine, searchable database, and GTM strategy with early pilot users

Research Assistant, Sunnybrook (Hospital) Research Institute – Toronto, Ontario July 2024 – June 2025

- Began as intern over the summer, hired back as RA under Dr. David Goertz in Focused Ultrasound (FUS) Lab
- Researched simultaneous optical and acoustic data of bubbles in agarose gel vessels for antivasular applications
- Developed a GUI for real-time signal processing and image segmenting, reducing manual analysis time by 80%
- Implemented **SOTA object tracking** models on micron-scale footage of bubbles in simulated blood vessels
- Employed new time-frequency transformations to visualize/analyze microbubble dynamics in FUS experiments

Founder, WOSS Robotics 8433 – Oakville, Ontario Sept 2023 – April 2025

- Founded and expanded a high school robotics program, leading 3 teams up to international competitions
- Raised over **\$14,000 in sponsorships**, collaborating with local businesses and STEM organizations
- Led the team to achieve **Rank #1** among Ontario high school robotics teams and a **Top 100** global skills ranking
- Numerous awards in Excellence, Skills, REC Challenges; qualified for **VEX World Championships 2025**
- Implemented outreach programs with local elementary schools to teach robotics, coding, and mechanical design and augmenting STEM curricula (recognized by school board for an **Inspire Award**), featured on **Oakville News**

Pharmacy Assistant, Mint Care Medical Centre – Oakville, Ontario Aug 2023 – Dec 2023

- Assisting pharmacists with medication dispensing and customer service, worked part-time for four months

Research

TraceGR: Modular Spacetime Imaging using Null Geodesics in Raycasting University of Toronto (2024)

 doi.org/10.1109/URTC65039.2024.10937597

Zane Beeai, Aditya Makkar, Dr. Ahmed Ellithy

Led research under the guidance of Dr. Ahmed Ellithy to create TraceGR: a null geodesic-based, modular raycaster to produce spacetime images rivaled only by Christopher Nolan in Interstellar. Presented at the **Canadian Undergraduate Math Conference (CUMC)** and **MIT-IEEE URTC**; peer-reviewed *IEEE Xplore* publication.

Projects

Solara CFC  /Solara_CFC

- Top 40 teams/56,000 participants in NASA Space Apps Challenge 2023. Pulled data from NASA & CSA satellites of CFC emissions, solar flux, and ozone to overlay on a mapping frontend, improving NASA's VISIONS software.
- Tools Used: Python, RASTER, JS, Flask, Next.js

WarRig X1 - Custom-built Electric Vehicle  /WarRig

- Designed and built 24V electric race car from scratch. Awarded Dennis Weishar Engineering Design Award at uWaterloo's EV Challenge. Funded by Town of Oakville for \$12,000 to make the car self-driving with LiDAR.
- Tools Used: CAD, FEA, Circuit Design, Python, LiDAR, Image Processing

Technologies

Languages: Java, Python, Javascript, C#, C++, VHDL, Visual Basic, R, Matlab

Technical Applications: Neural Networks (PyTorch, TensorFlow), RAG, GraphRAG, Signal Processing, CAS